

Introduction to Markov Decision Processes in Reinforcement Learning

Lesson 1 of RL for Data Science – Setting the Foundations



UCLA DSU W26 ML Curriculum

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Why Data Scientists Need RL

- Data science often focuses on *prediction*
- Real-world problems require **decision making** under uncertainty:
recommending products, A/B testing, or optimizing ad bids
- RL extends data science by learning **optimal actions** through **trial-and-error**:
interacting with environments like user behavior datasets or simulation models



Supervised versus Unsupervised Learning

Two Ways to Learn from Data



Supervised Learning

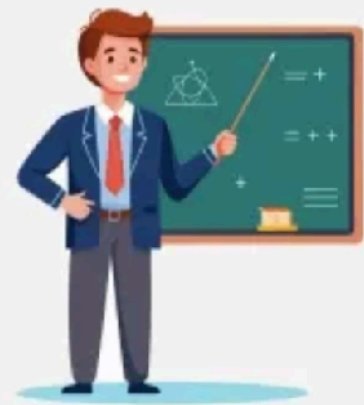
Learns from data with answers.



Unsupervised Learning

Finds hidden patterns in data without answers.

Supervised versus Unsupervised Learning



Supervised Learning

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Trained on
labeled data

Methods:
Classification
Regression



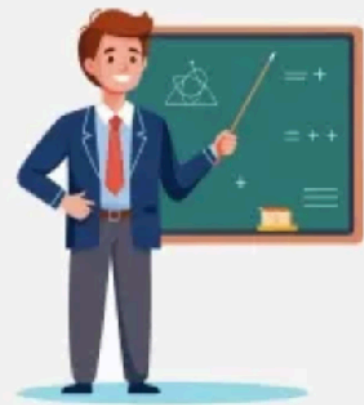
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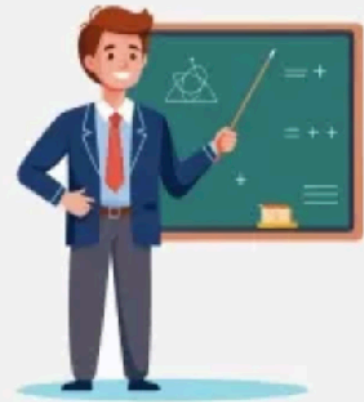
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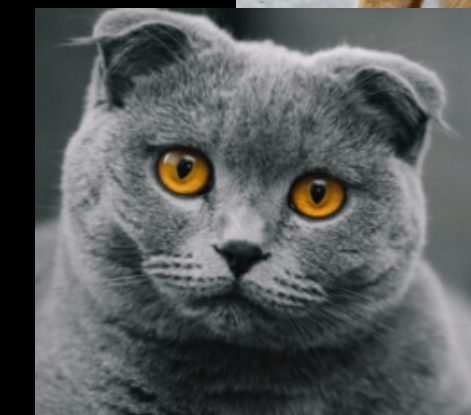
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Dog



Cat

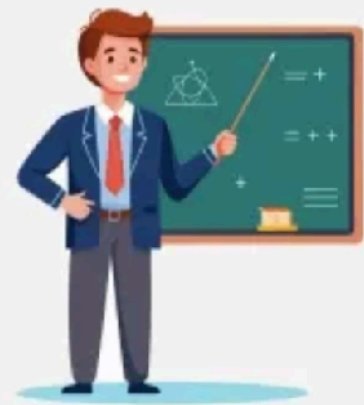


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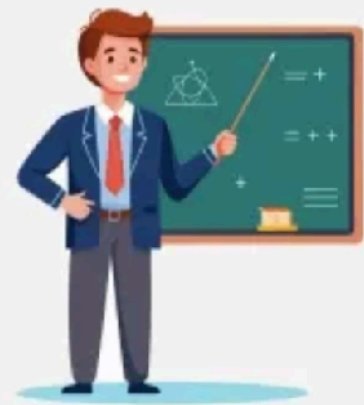
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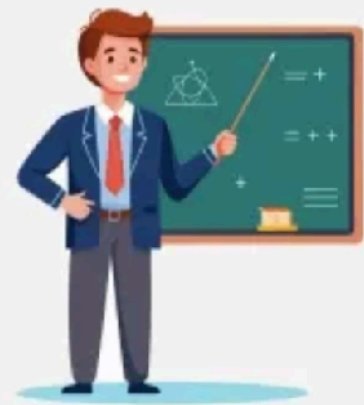
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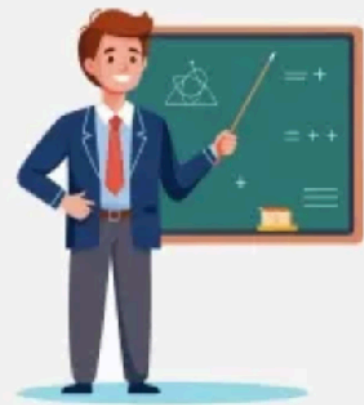
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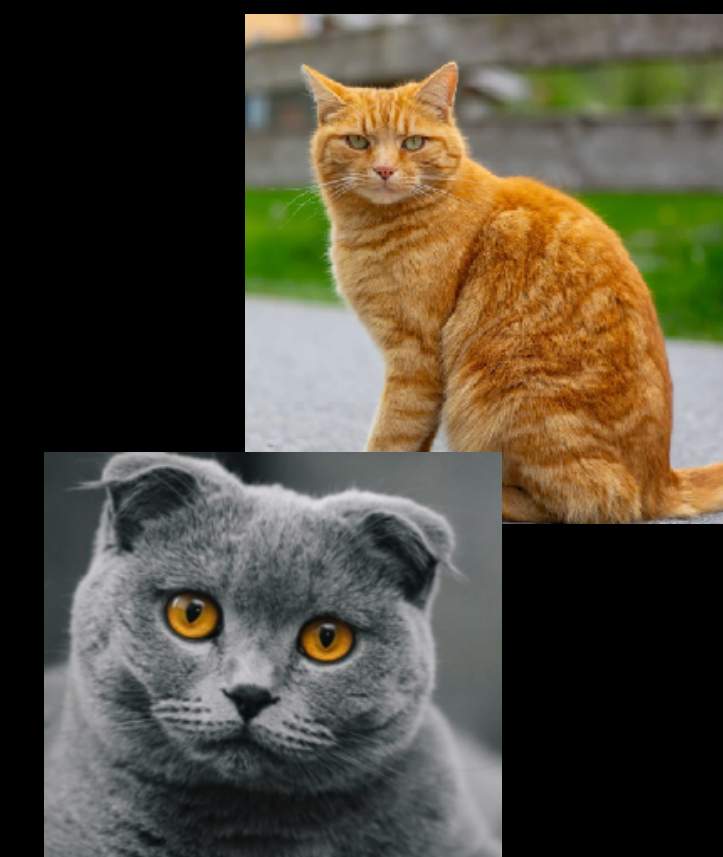
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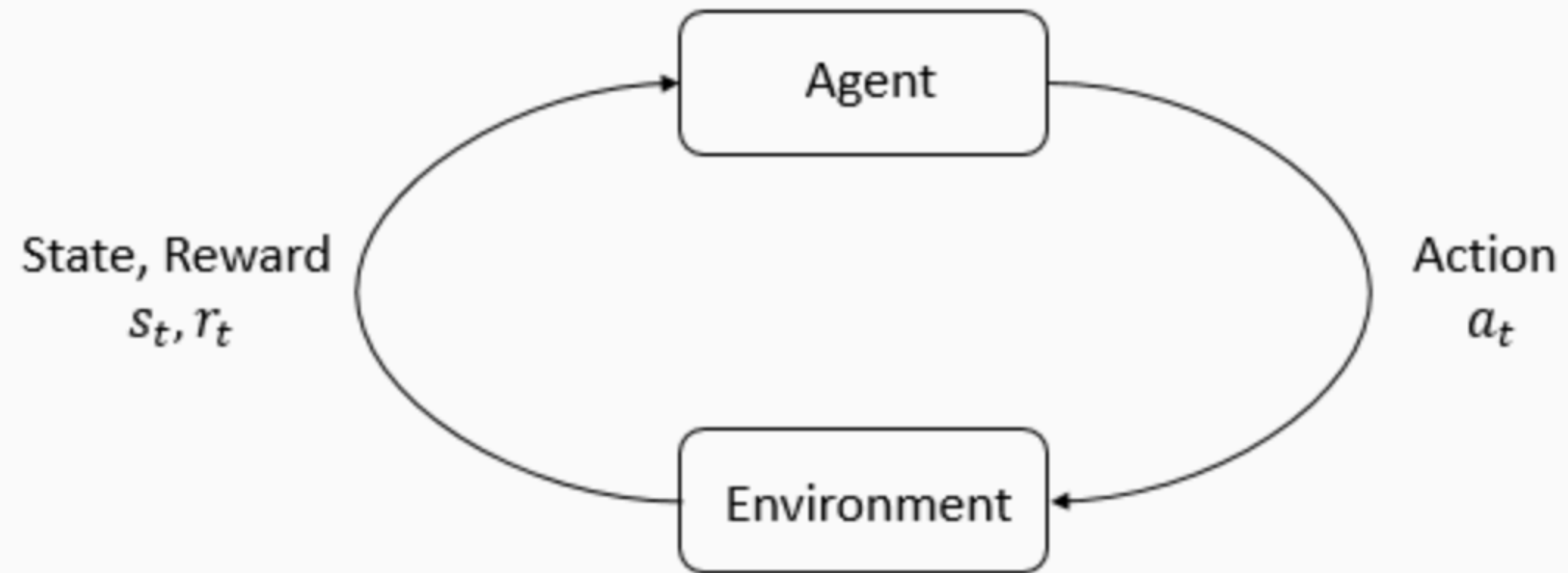
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RL as Unsupervised Learning

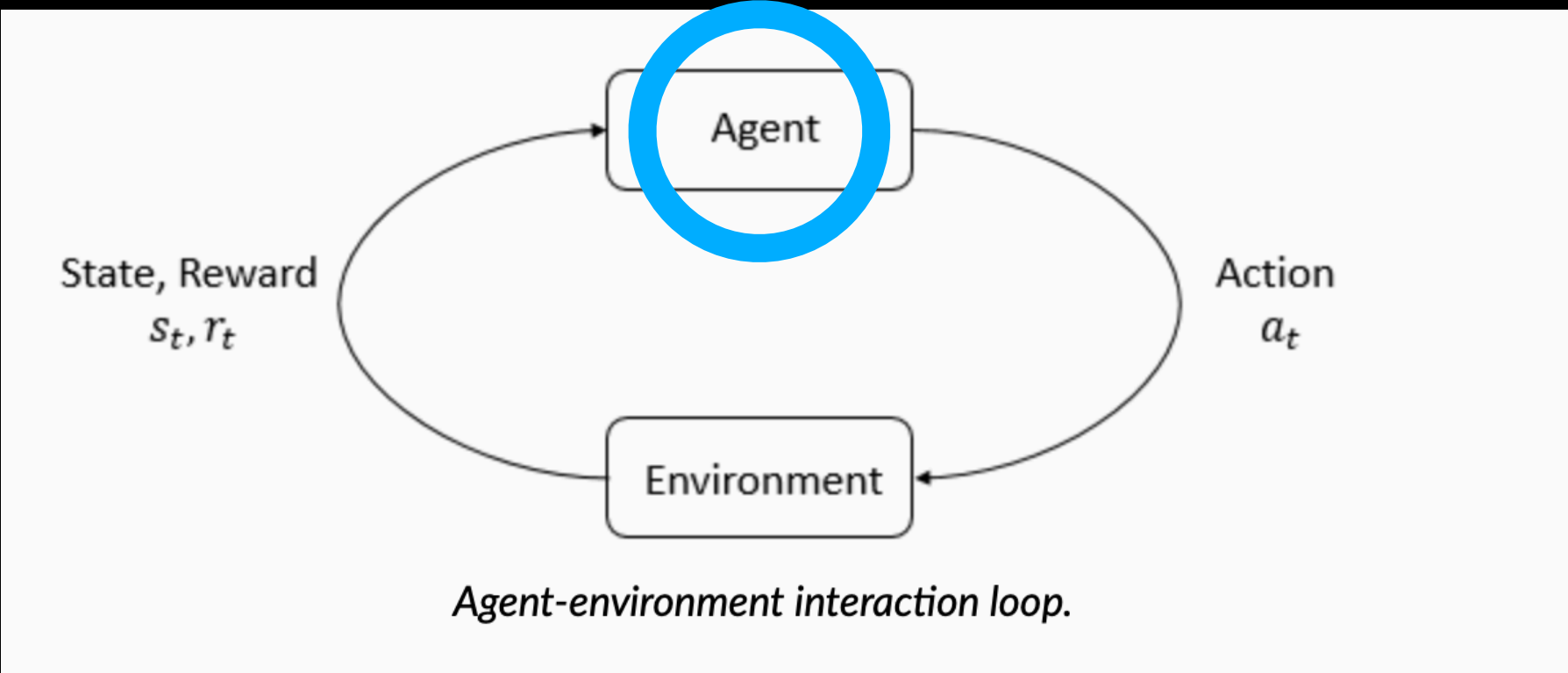
- No direct labels
- Learns from *rewards*
(optimization benchmarks),
dynamic environment, and
sequential directions
- RL isn't about *fitting* to known
answers, it's about *discovering*
what works best

Core RL Loop

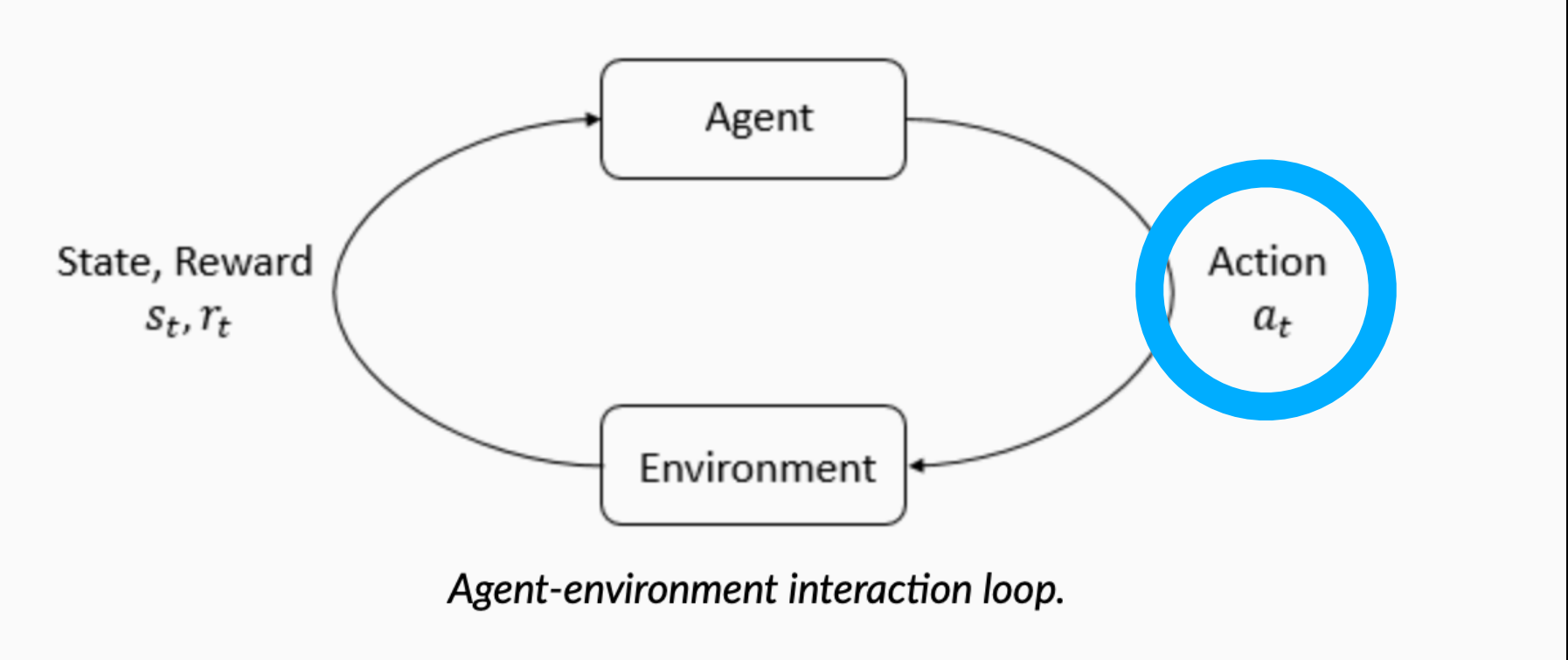


Agent-environment interaction loop.

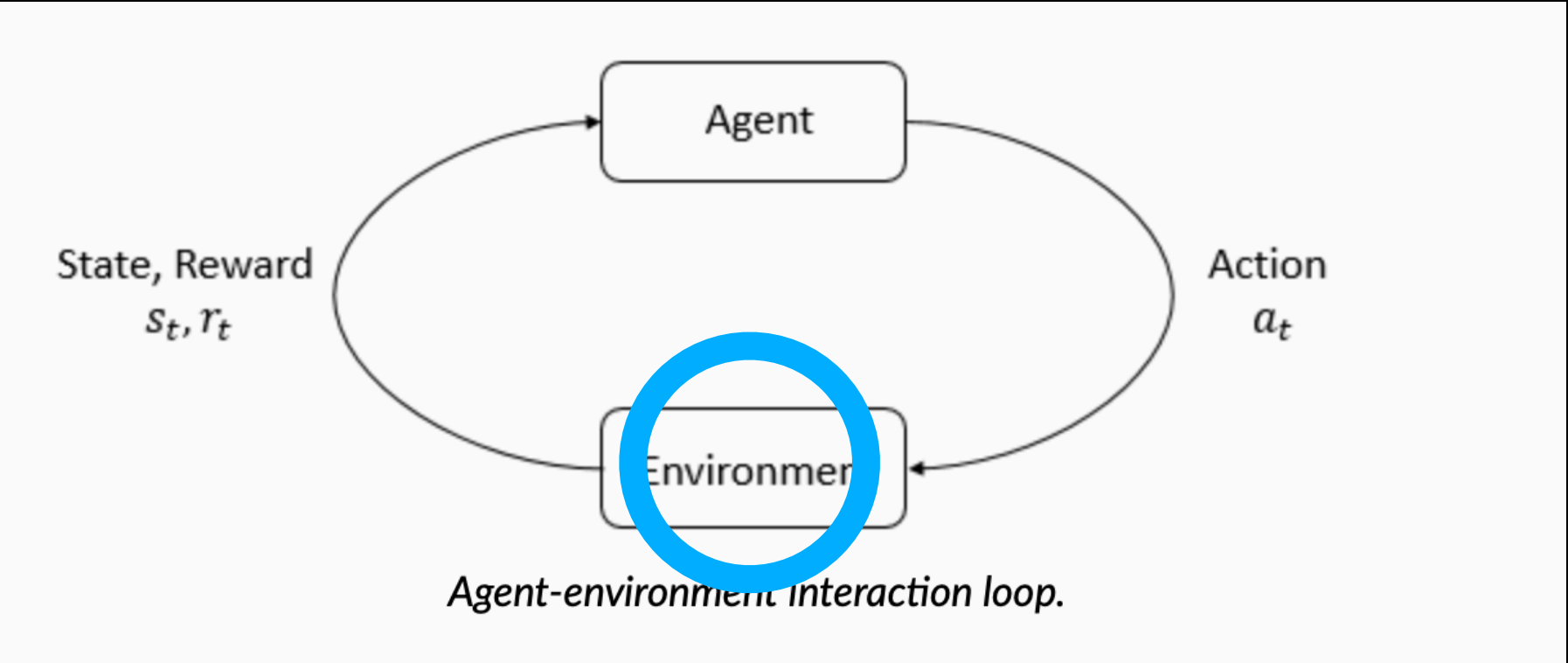
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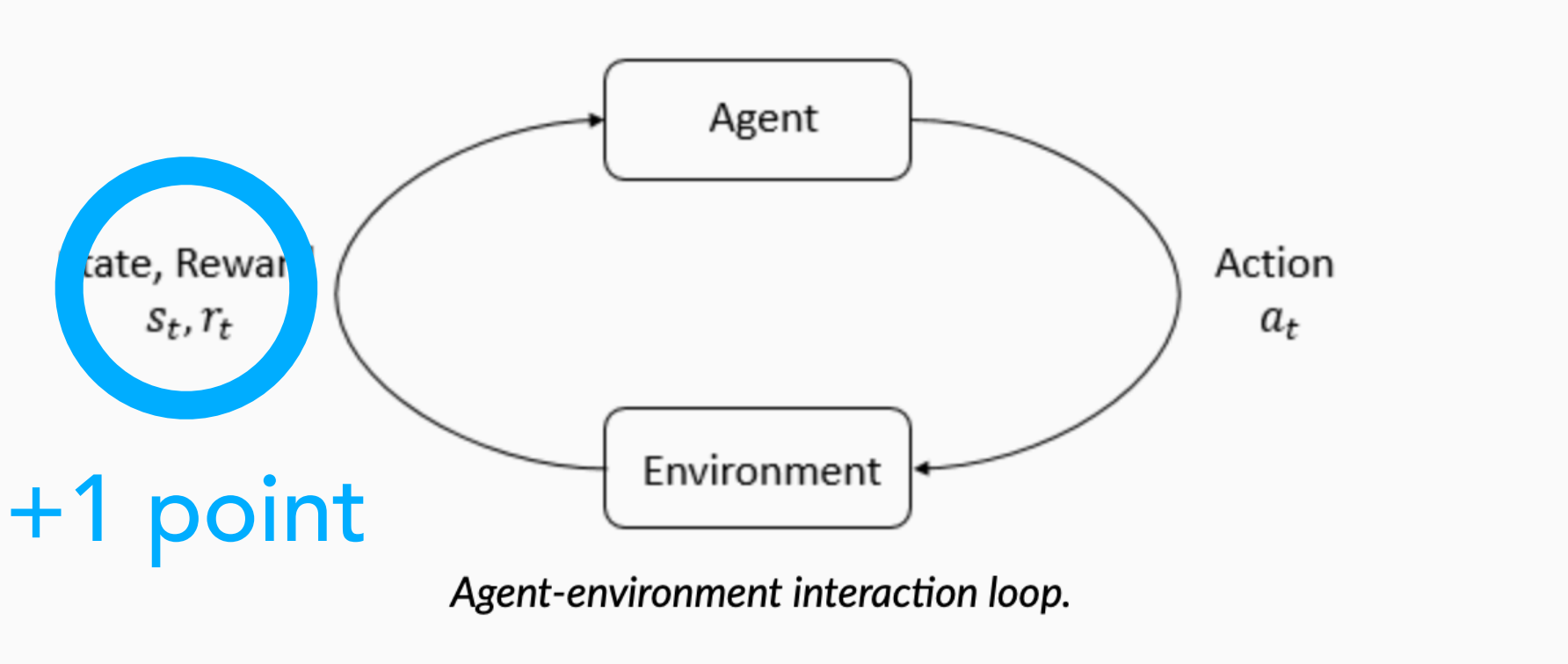
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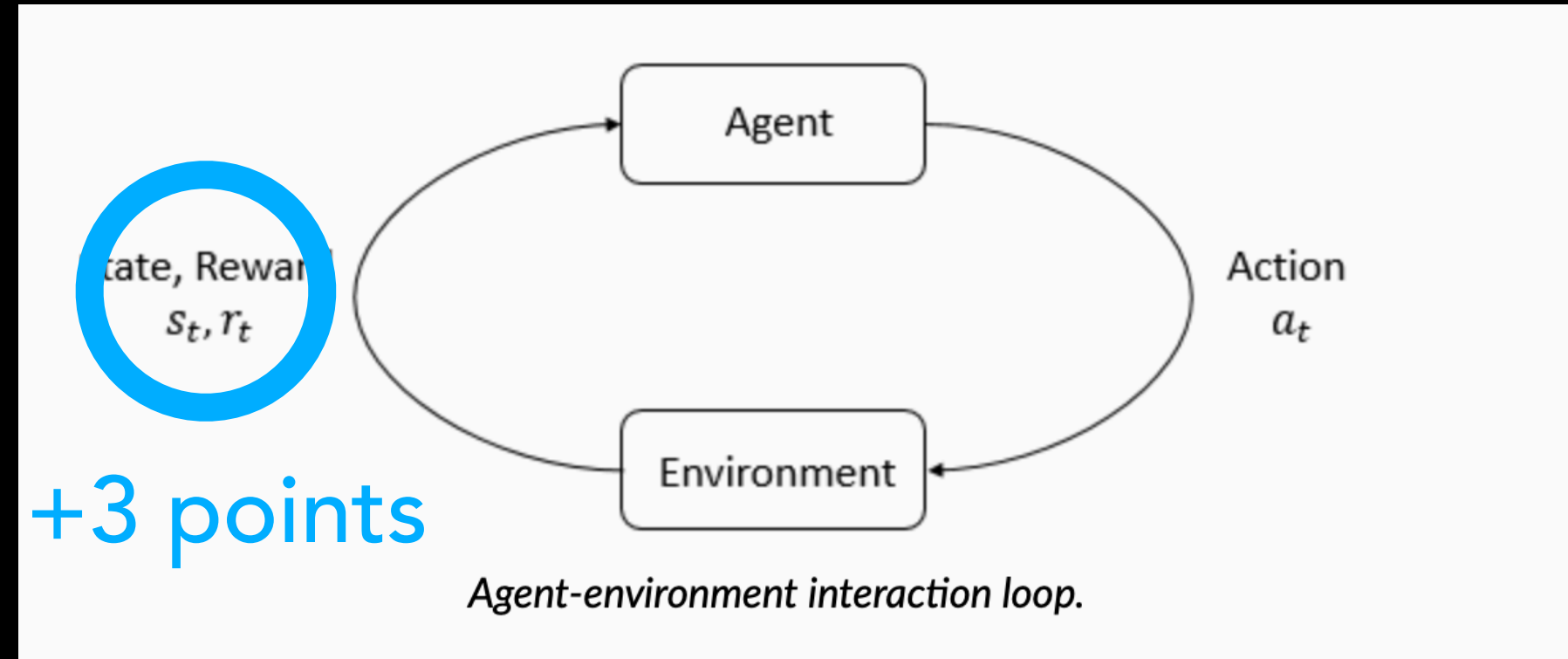
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Core RL Loop



Core RL Loop



- Note that the agent only **hypothesizes** the reward
- Reward is a **heuristic** set by the environment that aligns with the end goal



Questions?

Mathematical Foundations

How Does the **Agent** decide which **Action** is best?

$$\mathbb{E}[R] = \sum_{s'} P(s' | s, a) (R(s, a, s') + \gamma V(s'))$$

- **S**: State space (e.g., user session features in a rec system).
- **A**: Action space (e.g., items to recommend).
- **P**: Transition probabilities $P(s'|s,a)$ – next state given current state/action (Markov property: future depends only on present).
- **R**: Reward function $R(s,a,s')$ or $R(s,a)$ – immediate feedback.
- **γ** : Discount factor ($0 < \gamma < 1$) – balances short vs. long-term rewards (e.g., $\gamma=0.9$ in pricing to value future sales).

Deep Learning & Policies

- Now we have an agent/model that performs “intelligent” actions
- Now we need to train the model to improve its performance

$$\pi(a|s) = P(a|s)$$

- Policy: the **decision rule** on whether to take action a in state s
 - The optimal policy will maximize the longterm reward/win the game
- In Lesson 2 and Lesson 4, we'll explore ϵ -greedy and policy gradients

Q and Value Functions

- Q-Function $Q^\pi(s,a)$: Expected reward starting from s , taking a , then following π .
- Value Function $V^\pi(s)$: Expected cumulative reward starting from s , following π .
 - Relation: $V^\pi(s) = \max_a \{ Q^\pi(s,a) \}$
 - Here, we're trying to calculate the long-term reward as best as we can

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a, \pi \right]$$

$$V^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, \pi \right]$$

Background on the Homework

Homework: Navigating a FrozenLake Maze!

- For this week's assignment, we'll be building a model to navigate a "frozen lake" maze
- I'll provide a Jupyter notebook, and you'll fill in the exercises to build your model!

General Premise

- 4x4 grid, a starting and ending position with "frozen squares" and "hole squares"
- Goal is for the agent to get to the goal square without falling in a hole

Example Maze:

```
SFFF
FHFH
FFFH
HFFG
```


Homework: Navigating a FrozenLake Maze!

- **Environment:** 4×4 grid world from Gymnasium
- **States (S):** 16 discrete positions (0 to 15, row-major order: 0=top-left, 3=top-right, 15=bottom-right)
- **Actions (A):** 4 deterministic choices → 0=Left, 1=Down, 2=Right, 3=Up
- **Rewards (R):**
 - +1.0 only when you reach G (goal)
 - 0.0 everywhere else (including holes)
- **Episode ends** when you hit **G (success)** or **H (failure)**
- **Transition P:** In non-slippery mode → 100% you go where you intend (unless wall → stay put)
- **Goal of RL here:** Learn a policy that reaches G as often as possible (maximize expected return)